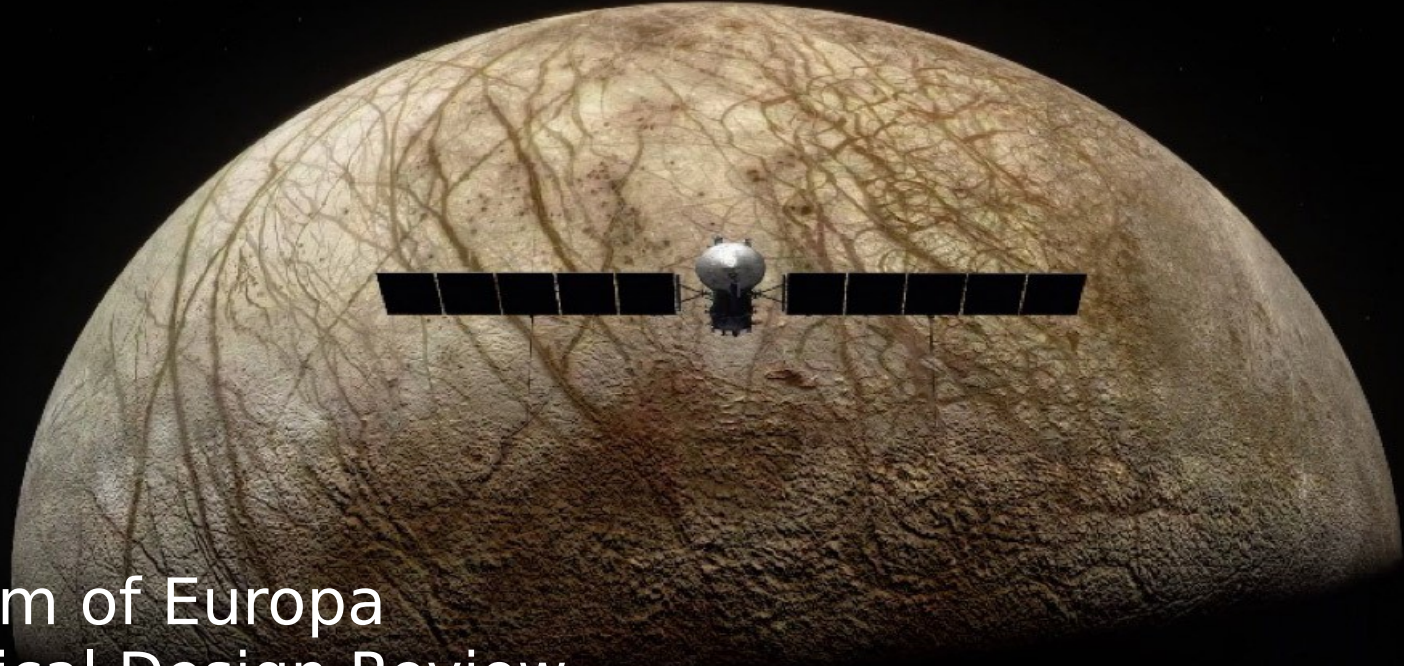




Pilum of Europa Critical Design Review

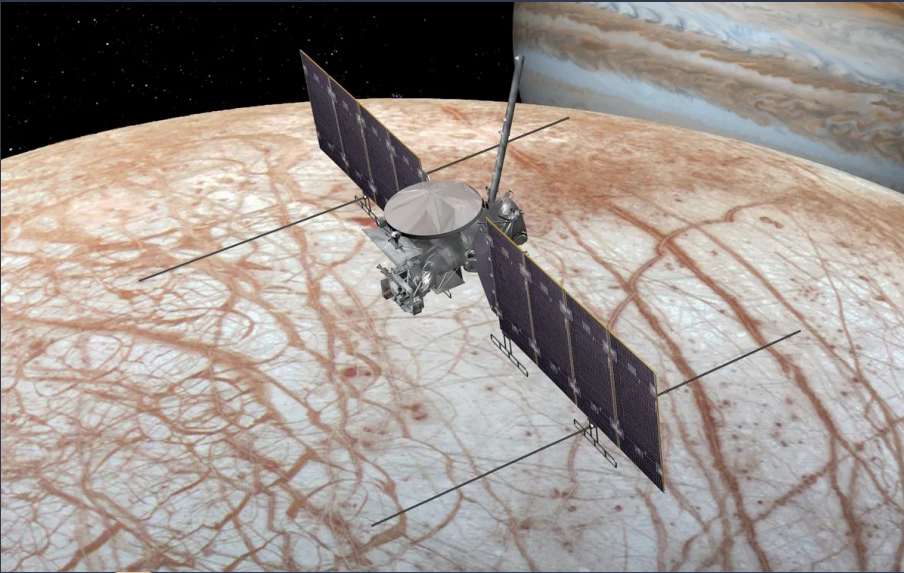
Michael Frank, Jack Morris, Jakob Schlosser, Thomas Todaro

Europa



Europa is the 6th closest moon of Jupiter with an orbital radius of $\sim 671,000$ km, period of ~ 84 hours, and is Jovian-synchronous, meaning, like our moon, the same half of Europa faces Jupiter at all times. Its size is about 90% of our moon's size meaning if gravitational acceleration is only 1.315m/s^2 . Its surface is covered in a thick layer, $\sim 15\text{-}25$ km of ice other than potential water vents that were seen by Hubble to be spewing liquid particulate into the space around it. Europa's extremely thin, oxygen rich atmosphere and presence of water, even in solid form, open up the possibility of discovering signs of life.

Mission Statement



Our mission would serve as a scientific reconnaissance mission to detect any signs of life, and more information about the ocean. Based on our findings we can figure out more about the past history of Europa. Judging by the water content and oxygen, we suspect organisms similar to primordial algae could exist in the vast oceans of ice.

This would consist of a deep space satellite with the objective of a high eccentricity, Jovian orbit in plane with Europa at periapsis to create a scan of its surface topography, determine future landing locations, and to drop a surface probe to drill/analyze surface samples and atmosphere. The spacecraft will also try to intercept one of these plumes of water to analyze the content from orbit, much like Cassini did on Saturn's moon Enceladus.

Upon completion of prior directives, we would like to use the satellite as a communications relay to transmit the data from the probe to earth. This mission will continue until sufficient surface data is recovered and nearly the entirety of Europa is scanned by the satellite. Following this we plan to use our leftover propellant to adjust orbits, observe Jupiter, and potentially other moons.

Objectives

Primary Objectives:

- Achieve a highly elliptical Jovian orbit with the periapsis at Europa's altitude.
- Deploy a probe to the surface of Europa to transmit surface sample and visual data.

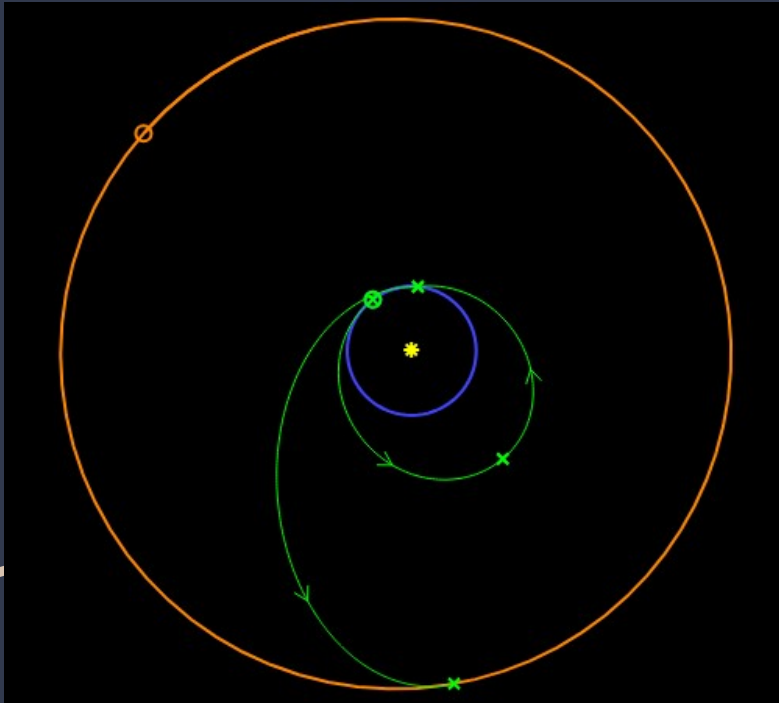
Secondary Objectives:

- Observe Jupiter and other moons
- Scout areas for future landings/exploration on Europa
- Intercept a water plume in orbit and collect samples from it for analysis
- Study radiation waves that come from Jupiter

Concepts of Operations- Orbital Mechanics

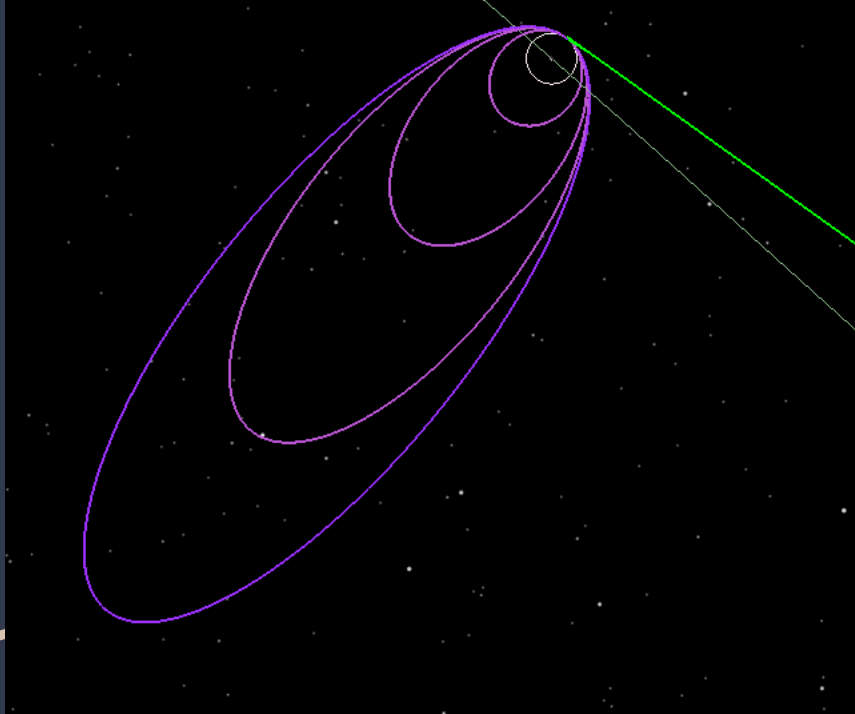


Interplanetary Travel



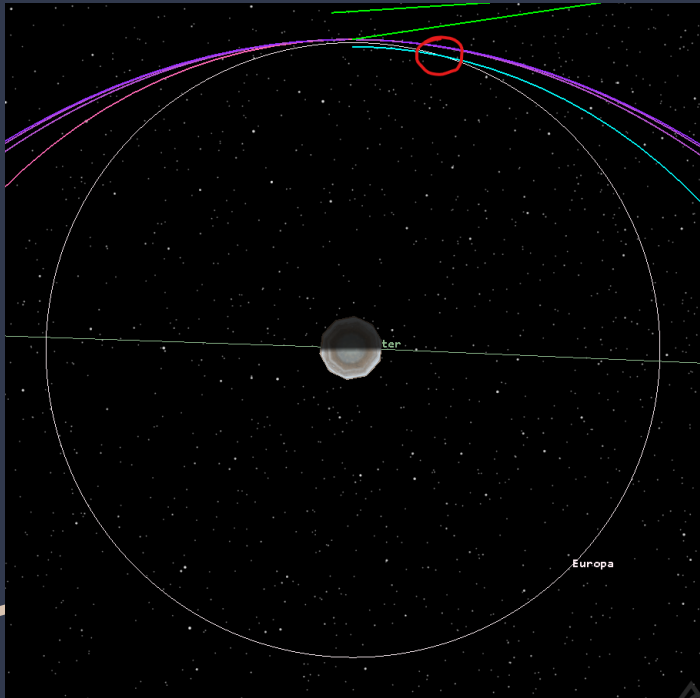
- After launch, Talos will circularize in a 200 km altitude parking orbit around Earth
- On Jan 27 2027, Talos will perform a 4.31 km/s maneuver to leave Earth SOI and raise its aphelion to prepare for an Earth gravity assist
- On Feb 1 2028, Talos will adjust the previous orbit to intercept Earth's SOI costing 0.511 km/s
- On Dec 15 2028, Talos will perform an Earth flyby costing 0.180 km/s but adding 13.38 km/s
- The Lambert transfer will then target Europa directly, catching the moon between Jupiter and the Sun, mid course corrections will cost ~ 0.222 km/s

Europa Injection



- Dec 8 2031- 2.93 years from gravity assist, Talos will enter Jupiter's sphere of influence
- Talos will inject into a highly elliptical orbit about Jupiter with its periapsis at altitude near Europa. This maneuver costs 815.583 m/sec
- After many gravity assisted, Europa fly-bys, the apoapsis will have been reduced to an eccentricity of about 0.5. We expect these maneuvers to cost us nearly ~ 150 m/s total in course adjustments over many months
- From here, Talos will scan Europa during its fly-bys and attempt to intercept a water plume

Deploying “Pilum”



- Pilum will initially be used in intercepting a ice crystal particulate cloud ejected from the surface of Europa.
- Following sufficient data collection, Talos will perform a maneuver to intercept Europa 5 days in advance costing 50 m/s.
- Talos will then decouple with Pilum and allow Pilum to collide with Europa
- Immediately after decoupling, Talos will perform another ~ 50 m/s maneuver to avoid collision with Europa.

Mass and Power Budgets



Mass Budget

Subsystem (Talos)	Mass Budget (kg)	Percentage
Science	146	7.4
Structures & Mechanisms	320.4	16.2
Thermal Controls	1.165	0.1
Power Generation	562.64	28.4
TT&C	150	7.6
On-board Processing	4.5	0.2
Guidance, Navigation, & Control	790	39.8
Propulsions	8	0.4
Total Mass	1982.705	100.0

Subsystem Pilum	Mass Budget (kg)	Percentage
Science	78.55	24.8
Structures & Mechanisms	100	31.6
Thermal Controls	0	0.0
Power Generation	46	14.5
TT&C	30	9.5
On-board Processing	61.69	19.5
Guidance, Navigation, & Control	0	0.0
Propulsions	0	0.0
Total Mass	316.24	100.0

Approximate mass of Pilum (Probe): 316.24 kg

Approximate dry mass of Talos (Satellite): 1982.705 kg

Estimated wet mass of Talos (including Pilum): 4281.65 kg

Power Budget

Subsystem (Talos)	Power Budget (Watts)	Percentage
Science	163.4	27.4
Structures & Mechanisms	9	1.5
Thermal Controls	18	3.0
Power Generation	6	1.0
TT&C	45	7.6
On-board Processing	30	5.0
Guidance, Navigation, & Control	319	53.6
Propulsions	5	0.8
Total Power	595.4	100.0
Power Generated	800	134.4
Subsystem Pilum	Power Budget (Watts)	Percentage
Science	117.4	49.9
Structures & Mechanisms	100	42.5
Thermal Controls	0	0.0
Power Generation	0	0.0
TT&C	15	6.4
On-board Processing	3	1.3
Guidance, Navigation, & Control	0	0.0
Propulsions	0	0.0
Total Power	235.4	100.0
Power Generated	125	53.1

The power system for Talos is extremely variable based on what devices are currently in use. We calculated for the most extreme power usage plus ~35% extra for redundancy and a wider range of options when retiring the satellite.

For the Pilum, the RTG we equipped it with is already very heavy in comparison to the rest of the probe so we have to be very conservative on power usage. No two science or mechanism systems can run at the same time but if activated in order and separated in procedure, the data collection and transmission should be successful.

Launch Details



NASA Space Launch System (SLS)



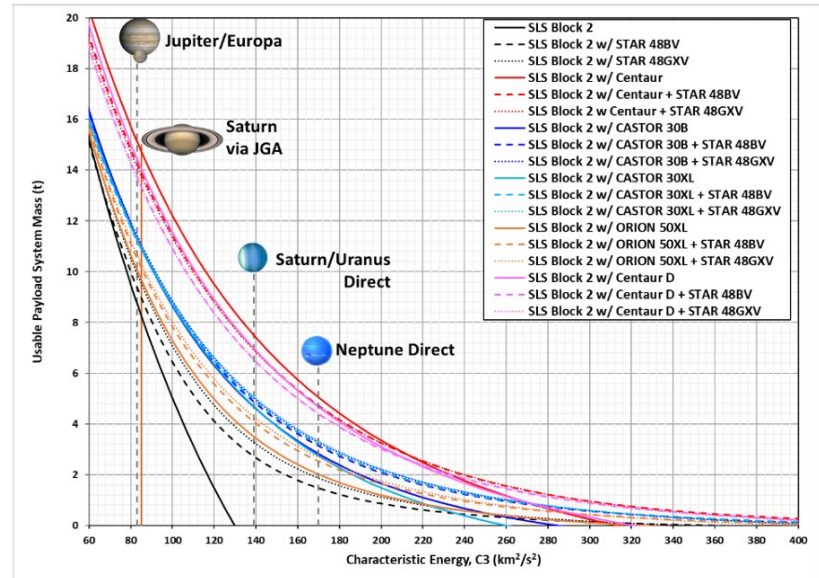
The NASA Space Launch System was designed to take large payloads into deep space and has a maximum launch thrust of 41 MN using block 2 configuration. The launch mass is 2600-2950 metric tons, with the propellant mass about ~50% of that. It can take about 6000 kgs to Jupiter on direct flight.

The SLS vehicle first stage uses 2 strap on solid rocket boosters in the first stage with Polybutadiene acrylonitrile (PBAN) solid propellant, with a specific impulse of 268s, along with a core booster with 4 RS-25D engines with LOX w/ LH2 propellant that has a specific impulse of 451s.

Launch Details



- Date: Jan 20 2027
- Launch Site: Cape Canaveral
- Inclination: 28.5 Degrees
- Require $C3 = 25 \text{ km}^2/\text{sec}^2$



Systems



Propulsion- Control Thrusters



- R-4D-11 Engine (Aerojet Rocketdyne) (2)
 - Bi-Propellant: 378-511N Thrust
 - Fuel: Monomethylhydrazine
 - Oxidizer: Nitrogen Tetroxide
 - ~4kg each nose cone
 - Used for small adjustments and the capture maneuver.
 - Second thruster for redundancy.
 - $I_{sp}=312$ seconds
 - Total Delta V: 2120.8 m/s

Guidance, Control, and Navigation

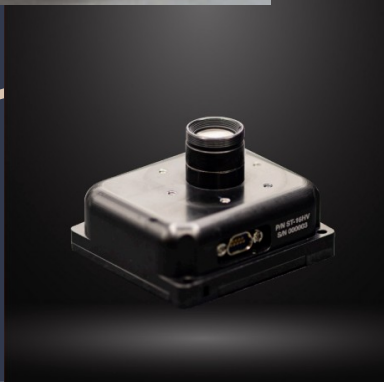


Reaction Wheels (4)

- HR16 by Honeywell
- Momentum Storage: 100 Nms
- Mass : 12kg each
- Power: 105W Max (<22W in Steady State)

Star Tracker & Attitude Tracker (2 each)

- ST-16RT2 by Rocket Lab
- Guides us near to Europa/makes sure our attitude is correct.
- Mass: 235 grams x 2 (need two for redundancy)
- Power: 1W x 2 of power
- ST-16HV by Rocket Lab uses data from tracker to determine attitude
- Mass: 148 grams x 2 (need two for redundancy)
- Power: 1W x 2 of power per



On-Board Processing



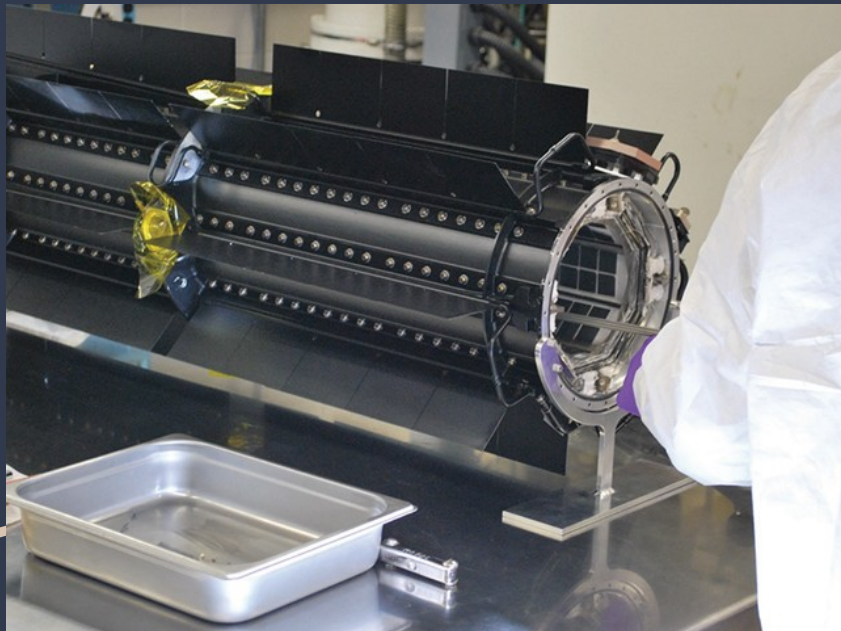
- SpaceCloud® iX5-106 (Talos)
 - Weight: 4.5kgs
 - Power Consumption: 10-30W depending on usage.
 - Storage: 2 x 120 GB SATA SSD.
- Used for On Board Processing along with data storage and communication.
- Will handle all communication between devices and computing power for the devices on board.
- Kryten-M3 (Pilum)
 - Weight: 61.69 grams
 - Power Consumption: 400 mW - 1W max
 - Storage: 4 GB
- All Are Radiation Hardened/Resistant

Telemetry, Tracking, & Communication



- The satellite will use a X band radio frequency of 8.4 GHz and a Ka band radio frequency of 32 GHz.
 - Probe will use UHF band radio frequency of 3 GHz.
 - Frequency Band Selection approved by The National Telecommunications and Information Administration (NTIA).
 - The modulation technique for both the uplink and downlink will use Quadrature Phase-Shift Keying (QPSK) to support high data rates, the best power efficiency possible, and relatively narrow main lobe width
 - A 3m diameter High gain to High gain antenna for satellite
 - Low and high gain antennas on probe.
-
- Peak Operating Power High Gain Antenna: 30 W
 - Mass of High Gain Antenna: 120 kg
 - Peak Operating Power Low Gain Antenna: 15 W (2, one for Pilum)
 - Mass of Low Gain Antenna: 30 kg (2, one for Pilum)

Power Generation and Storage



Talos:

- Power Generation:
 - Triple Junction Gallium Arsenide Array (Ultra MJ)
 - Industry standard, high collection efficiency, and suitable for areas of higher radiation, such as the space around Jupiter
 - Required Area: 58.05 m^2
 - Power: +773.76 W (compensating for eclipse times)
 - Mass: 162.55 kg
 - GPHS-RTG (General-Purpose Heat Source- Radioisotope thermoelectric generator)
 - We will need 2 units
 - Each unit:
 - Power: +300 Watts
 - Mass: 57 kg
 - Volume: 0.422m Diameter x 1.14 m Long
 - Lifetime: 14 years
- Power Storage:
 - Lithium Ion Batteries
 - Extremely efficient, very little loss
 - Must store ~575 Watts while sun exposed
 - Mass: 4.59 kg
 - Talos Power Total:
 - Power: 800 W average
 - Mass: 281.14 kg

Pilum:

- Power Generation
 - MM(Multi Mission)-RTG
 - Power: +125 W
 - Mass: 45 kg
 - Volume: 0.64 m Diameter x 0.66 m tall
- Power Storage
 - Lithium Ion Batteries
 - Mass: 1kg
 - Redundant power storage

Thermal Regulation



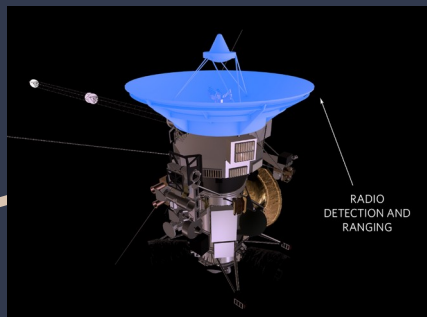
Insulation: Space Grade Mylar Sheets

- Area = 5 m x 1.2 m
- Weight = 1 kg
- Composed of aluminized Mylar and glass fabric
- Can insulate temperatures between -250°F and 250°F

Heater: designed to maintain the surface temperature of the manned spacecraft body and the fastening elements of devices and assemblies within a given temperature range (3 units)

- Weight = 0.055 kg each
 - 0.165 kg total
- Power = 5.897 W each
 - 17.691 W total

Science Instrumentation



Talos:

- Multispectral Imaging System
 - Wide angle camera and narrow angle camera detecting visible light and some small infrared and ultraviolet wavelengths.
 - Mass: 58 kg
 - Peak Operating Power: 22 W
 - Peak Data Rate: ~365 kilobits/sec
 - Size: 95x40x33 cm (NAC) 55x35x33 cm (WAC)
- Synthetic Aperture Radar System
 - Radio waves to map the topography of Europa
 - Altimeter and Radiometer
 - Mass: 45 kg
 - Peak Operating Power: 109 W
 - Peak Data Rate: ~364.8 365 kilobits/sec
 - Size: 16 m- retractable
- 3 Sensor Magnetometer
 - Reads in magnetic field information from nearby bodies
 - Mass: 31 kg
 - Power: 16.2 W
 - Data Bit Rate: 3.60 kilobits/sec
 - Size: 8 meter collapsable boom
- Plasma Instrumentation Systems
 - Magnetic field near Europa carries charged particles called plasma that distorts the magnetic field and its signal.
 - Mass: 12 kg
 - Peak Operating Power: 16.2 W
 - Size: 4 Faraday cups each 7 cm deep and 20 cm wide

Total Mass: 146 kg

Total Power: 163.4 W

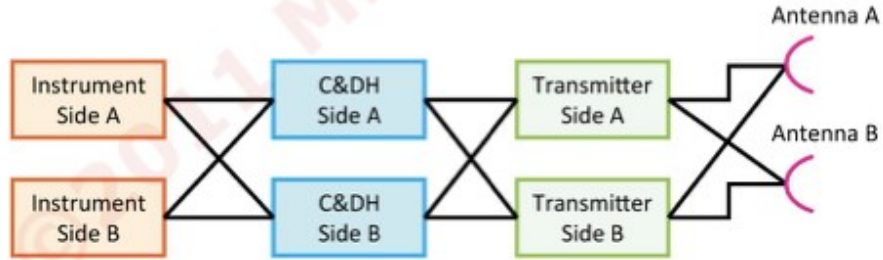
Pilum

- Mass Spectrometer
 - Analyzes and determines the content of gases exposed to it
 - Mass: 61.75 kg
 - Power: 97 W
 - Size: 0.268 x 0.250 x 0.171 m
- Surface/Dust Sample Analyzer
 - Analyzes and identifies small, solid matter
 - Mass: 16.8 kg
 - Peak Operating Power: 20.4 W
 - Size: 0.268 x 0.250 x 0.171 m

Total Mass: 78.55 kg

Total Power: 117.4W

Electrical Structure



- The structure we are going for is a block redundant and cross-strapped configuration.
- Due to the duration length of the mission and the high amount of radiation coming from Jupiter, an electronics failure is more likely to happen.
- This will double the amount of components and wiring but will greatly increase the success rate.

Spacecraft Design



Structure & Mechanisms



Structure

- We will use an aluminum 7075 frame because of its strength and anti-corrosion properties.
- Density: 2.81 g/cm³.
- Mass ~250-300 kg

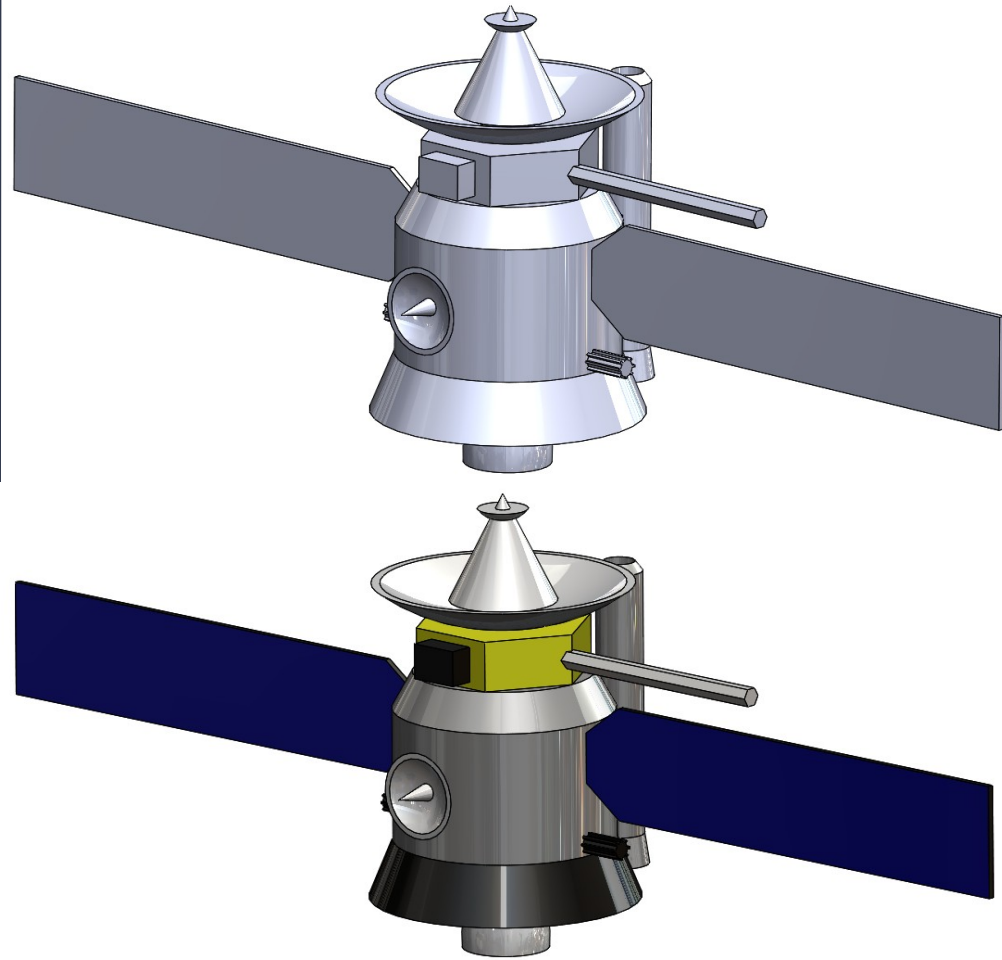
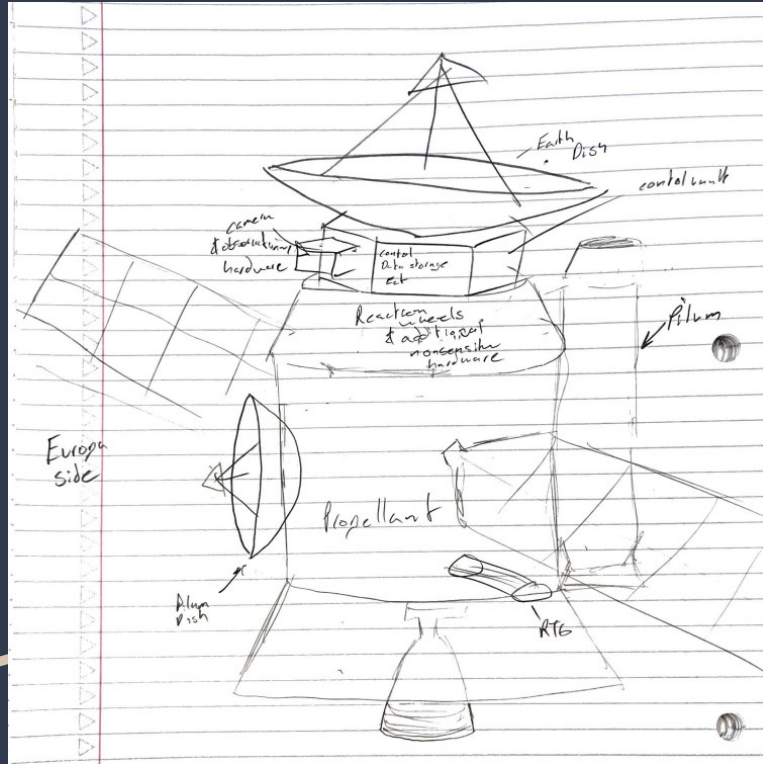
Talos Mechanisms

- Pilum Release Motor:
 - Power: 3 W
 - Mass: 1.6 kg
- Solar Array System Gimbels (2):
 - Power: 3 W each
 - 9.4 kg each

Pilum Drill: Specifically designed for this mission:

- Low power usage: <100 W
- Low mass: <10 kg
- Long drill times
- High torque

Preliminary Sketch & 3D Design



Sources

- <https://science.nasa.gov/mission/cassini/spacecraft/cassini-orbiter/magnetometer/>
- https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/Cassini/instant-mag.html
- https://en.wikipedia.org/wiki/Radar_for_Europa_Assessment_and_Sounding:_Ocean_to_Near-surface
- https://en.wikipedia.org/wiki/Multi-mission_radioisotope_thermoelectric_generator
- <https://en.wikipedia.org/wiki/GPHS-RTG>
- <https://europa.nasa.gov/mission-updates/41/10-things-to-know-about-europa-clipper-science-instruments/>
- <https://meetingorganizer.copernicus.org/EPSC2014/EPSC2014-229.pdf&lang=en>
- https://europa.nasa.gov/internal_resources/379/ScienceInstruments_031422_Public.pdf
- https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Onboard_Computers_and_Data_Handling/